

Deep Vaviya

Ghatkopar (W), Mumbai, India
+91 9004946606 | deepvaviya16@gmail.com
[Portfolio](#) | [LinkedIn](#) | [GitHub](#)

Summary

Computer Engineering student (B.Tech 2027) with hands-on experience in full-stack development, AI/ML, and intelligent software systems. Skilled in building scalable web applications using React, Node.js, RestAPI, and MongoDB, while developing AI-powered solutions involving computer vision, deep learning, and real-time signal processing. Strong foundation in algorithms, REST APIs, and modern software engineering practices with experience delivering production-ready applications through internships and hackathons.

Technical Skills

- **Languages:** Python, Java, JavaScript, C++, SQL, PHP
- **AI/ML:** Deep Learning, rPPG Modeling, Signal Processing, Computer Vision, Physiological Analytics
- **Libraries:** OpenCV, MediaPipe, NumPy, SciPy, FFT, Butterworth Filters
- **Frontend:** React, Bootstrap, HTML, CSS
- **Backend:** Node.js, Express.js, FastAPI, REST APIs, WebSockets
- **Databases:** MongoDB, MySQL
- **Core CS:** Data Structures, Algorithms, OOP
- **Tools:** Git, GitHub, Postman, VS Code

Projects

VitalSense — Contactless Vital Signs Monitor *Python, FastAPI, React*
github.com/DeepVaviya/Vital-Sense

- Built real-time contactless physiological monitoring system using webcam-based rPPG.
- Designed deep learning rPPG signal extraction pipeline using POS + CHROM fusion.
- Implemented Butterworth filters, FFT, and Lomb-Scargle for HR and HRV estimation.
- Computed HR, RMSSD, SDNN, and LF/HF ratio.
- Developed AI-based stress, fatigue, and mood prediction.
- Built FastAPI + WebSockets backend with real-time React dashboard.

Voice Trace — AI Audio Processing & Financial Tracking *React, Node.js, AI*
voicetracee.netlify.app

- Built Whisper-powered voice transcription and audio analysis platform.
- Engineered secure backend with automated webhooks and smart processing flows.
- Developed Smart Insights engine for extracting financial trends from voice inputs.
- Created automated ledger management system for categorized entries.
- Integrated loan analysis and weather services for context-aware recommendations.
- Designed scalable React frontend with Node.js/Express backend architecture.

Portfolio Website *HTML, CSS, JavaScript*
deepvaviya.netlify.app

- Designed and deployed responsive portfolio showcasing projects and technical skills.

Experience

Software Development Intern — BZTech Pvt. Ltd., Mumbai *Dec 2025 – May 2026*

- Developed full-stack web applications and REST APIs for production use.
- Improved application performance and optimized database operations.
- Collaborated using Git workflows, debugging, and feature deployment.

Education

- **B.Tech in Computer Engineering**, Shah and Anchor Engineering College
CGPA: 8.82 2024 – 2027
- **Diploma in Computer Engineering**, Vidyalankar Polytechnic
89.88% 2024
- **SSC, M.D. Bhatia Secondary School**
82.20% 2021

Achievements

- Winner – Best UI/UX (Rs.10,000 Prize) at Hackverse 2.0 National Hackathon (2026)
- Strong hackathon-driven development experience with rapid prototyping.

Additional

- **Languages:** English, Hindi, Gujarati, Marathi
- **Interests:** AI Systems, Computer Vision, Real-time Analytics, Scalable Software